

Acute Liver Failure (ALF) — Clinical One-Pager

Facilitator quick reference · TeachIM (Mowry/Kriss, Apr 2022), optimized Jun 2026 · CC BY-NC 4.0 · clinician-review before use

ALF = ALL three: acute liver injury (<26 wk, no chronic liver dz*) + coagulopathy (INR >1.5) + hepatic encephalopathy. **HE is what makes it "failure."** *exc: HBV, AIH, Wilson's.

Definition & HE Grades

Criterion	Detail
Injury	↑AST/ALT, bili; <26 wk; no chronic dz
Coagulopathy	INR >1.5 (key prognostic marker)
Encephalopathy	West Haven I–IV; defines ALF

HE grades: I sleep-wake/attention · II lethargy+asterixis · III stupor/gross disorient · IV coma. **Gr III–IV → intubate.**

Etiologies & Work-up

Category	Causes / work-up
Drugs/toxins (~50%)	APAP (lead), supplements, Amanita; statins/AEDs/abx. APAP+salicylate lvl, utox, LiverTox
Viral	Hep A/B/D/E; rare HSV/VZV. Serologies+PCR, HIV
Ischemic/vascular	Budd-Chiari, shock liver. Liver US doppler
Other	AIH, preg (HELLP/AFLP), Wilson's. ANA/ASMA/IgG, preg test, ceruloplasmin+24h Cu, bx

Alcohol does NOT cause ALF (can worsen another insult).
Wilson's clue: ALP:Tbili <4, young, Coombs-neg hemolysis.

ICU Management (4 steps)

1. Complications:

Problem	Action
Cerebral edema	q1-2h neuro checks; HOB>30; hypertonic saline/mannitol; MINIMIZE IVF ; NH3 >150-200 = high ICH risk
Hypoglycemia	q1-2h glucose; dextrose infusion
MOF	Vent/pressors/RRT; TTE r/o cardiogenic
Coagulopathy	Vit K ok; FFP/factors ONLY if bleeding
Infection	High risk; consider empiric abx/antivirals

2. NAC empirically in ALL ALF (benefit even non-APAP, best early/low-grade HE). **3. Prognosticate** KCC (±MELD) w/ hepatology. **4. TRANSFER** early to LT center.

King's College Criteria

APAP ALF	Non-APAP ALF
pH <7.3 OR all: INR >6.5, Cr >3.3, gr III–IV HE	INR >6.5 OR ≥3 of: age <10/>40; HAV/HBV/drug; jaundice >7d before HE; INR >3.5; Tbili >18

Pearls

- **No HE = acute liver INJURY, not failure.**
- Don't reflexively correct INR — it's your prognostic compass.
- Go easy on IVF even with rising Cr — volume worsens cerebral edema.
- KCC is specific, not sensitive — a negative KCC should NOT defer transfer.
- Transfer EARLY, before progression precludes safe transport.